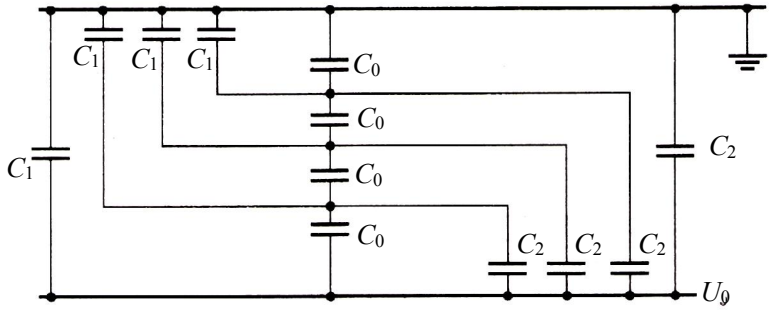
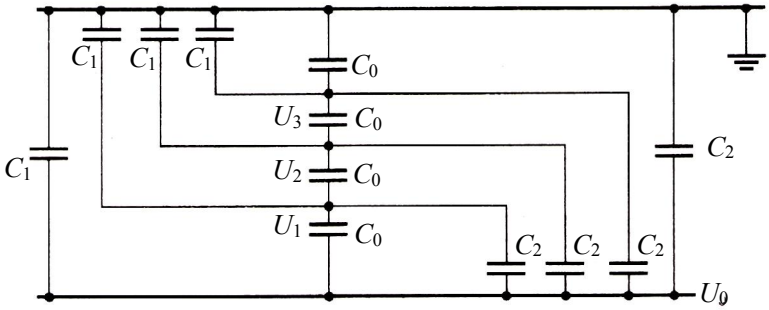


3. Let the intermediate potentials be U_1 , U_2 , U_3 as shown. At each node, the algebraic sum of charges on the four connected capacitor plates is zero:

$$\begin{cases}(U_1-U_2)C_0+U_1C_1-(U_0-U_1)C_0-(U_0-U_1)C_2=0,\\(U_2-U_3)C_0+U_2C_1-(U_1-U_2)C_0-(U_0-U_2)C_2=0,\\U_3C_0+U_3C_1-(U_2-U_3)C_0-(U_0-U_3)C_2=0.\end{cases} \tag{9}$$



The four insulator voltages sum to U_0; thus

$$(U_0-U_1)+(U_1-U_2)+(U_2-U_3)+U_3=U_0\ . \tag{10}$$

Define

$$\triangle U_1=U_0-U_1\ ,\quad \triangle U_2=U_1-U_2\ ,\ \triangle U_3=U_2-U_3\ ,\ \triangle U_4=U_3\ , \tag{11}$$

Rearranging (9) gives

$$\begin{cases}\triangle U_1(C_0+C_1+C_2)-\triangle U_2C_0-U_0C_1=0,\\\triangle U_1(C_1+C_2)+\triangle U_2(C_0+C_1+C_2)-\triangle U_3C_0-U_0C_1=0,\\\triangle U_1(C_0+C_1+C_2)+\triangle U_2(C_0+C_1+C_2)+\triangle U_3(2C_0+C_1+C_2)-U_0(C_0+C_1)=0;\end{cases}$$

Substituting the data gives

$$\begin{cases}76\triangle U_1-70\triangle U_2-5U_0=0,\\6\triangle U_1+76\triangle U_2-70\triangle U_3-5U_0=0,\\76\triangle U_1+76\triangle U_2+146\triangle U_3-75U_0=0.\end{cases} \tag{12}$$

Solving (12) gives

$$\triangle U_1=0.298\ U_0\ ,\quad \triangle U_2=0.252\ U_0\ ,\ \triangle U_3=0.228\ U_0\ . \tag{13}$$